Anthony Petrescu

Eden Prairie, Minnesota 55347 | anthonypetrescu@gmail.com | linkedin.com/in/anthony-petrescu | 9anthony0.github.io

EDUCATION

Purdue University | West Lafayette, Indiana Graduate: May 2027
Finance and Computer Science (Machine Learning Track)
Honors/Awards: Dean's List and Semester Honors
Clubs/Extracurriculars: Honors Research Accelerator LC, Executives in Residence, SigmaPi Fraternity, Latin Ballroom Dance Team

---

TECHNICAL SKILLS

Programming: Java, Lua, Python, C, C++, R-Studio, SQL
Machine Learning & Data: PyTorch, Scikit-Learn, Pandas, Numpy, RAG Workflows, Monte Carlo Simulations
Frameworks & Tools: Streamlit, Plotly, Asyncio, Git, CI/CD Principles, REST APIs, proficient with MS 365 Applications

---

PROFESSIONAL EXPERIENCE

Trane Technologies *Technical Sales - Controls Intern* | Arden Hills, Minnesota May – Aug 2025

- Crafted estimates and specification proposals for controls jobs with valuations of up to $85,000
- Analyzed existing customers/accounts, including drafting a SWOT analysis, to create 15+ job opportunities via SalesForce
- Created and priced retrofit (upgraded) controls solutions for customers' legacy control systems, valuing north of $35,000

Vertically Integrated Projects *IoT & Machine Learning Engineer* | West Lafayette, Indiana Jan – May 2023

- Leveraged IoT devices to collect a diverse range of data from sensors that measure 13+ environmental variables
- Applied advanced machine learning algorithms to simulated hazard scenarios, creating a predictive system
- Presented team's findings/solution at Purdue's 2024 Spring Undergraduate Research Conference

EAM-Mosca *Intern* | Hazle, Pennsylvania Aug – Oct 2022

- Optimized Sonotrode placement, increasing packaging equipment efficiency
- Designed and implemented Excel-based tracking systems for Sonotrode data and packaging performance metrics

---

PROJECTS

Intraoperative Surgical Phase & Gesture Recognition Engine (Python, PyTorch, Streamlit)

- Architected an attention-augmented causal Temporal Convolutional Network (TCN) in PyTorch to process simulated 8-DOF robotic surgical kinematics at 30 Hz**.**
- Engineered a low-latency streaming inference pipeline using asyncio and fixed-capacity ring buffers, achieving a p99 end-to-end latency of ~3.1 ms (16x faster than the 50 ms budget constraint).
- Built a clinical-grade "Surgeon Console" dashboard using Streamlit, featuring real-time 3D instrument trajectory tracking, EMA-smoothed anomaly probabilities, and hardware-style LED alert hysteresis.
- Designed the model with separate Multi-Task classification heads to independently detect surgical gestures (Suturing, Dissection) and anomalies (Micro-Hesitations, Tremors).

Dynamic Portfolio Optimization Engine (Python)

- Developed a live Portfolio Optimizer in Python that ingests real-time market data to calculate covariance matrices and visualize the Efficient Frontier, allowing users to dynamically stress-test allocations for maximum risk-adjusted returns.

Dynamic DCF Valuation Engine (Python)

- Developed a live, interactive Discounted Cash Flow (DCF) engine in Python that bridges Enterprise to Equity value and features dynamic 5x5 sensitivity matrices to stress-test intrinsic valuations against live market prices.

Golf Tournament Predictive Engine (Python)

- Developed a professional-grade predictive engine in Python that ingests player statistics to evaluate complex feature interactions, mathematically mapping the exact probability of tournament victories based on course-specific skill demands.

World Cup Predictive Engine (Python)

- Correctly predicted the final and winner by developing a Monte Carlo simulation engine that ingests squad depth metrics and real-time form indicators to run 10,000 tournament iterations, applying exponential Elo scaling and custom penalty damping logic to mathematically map tail-risk upset probabilities.

LEADERSHIP & OTHER EXPERIENCE

Cary Quad Resident Assistant | West Lafayette, Indiana Aug 2024 – Present

- Managed a $96,000 budget to coordinate 36+ hall-wide events, promoting student engagement, personal development, and health resources for residents while cultivating an inclusive environment for residents from diverse backgrounds

Boy Scouts of America Troop 347 *Eagle Scout and Patrol Leader* | Eden Prairie, Minnesota Feb 2018 – Present

- Led a team of 15 scouts in planning and executing a Christmas tree lot, raising over $5,000 for a local church
- Organized a team of 20 volunteers to execute a cybersecurity awareness Eagle Project for the Eden Prairie Senior Center